WORKED SOLUTION

STEP III 1998, Question 10

Mechanics • Oblique Collisions

STEP III 1998, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q10

The key idea

With equal masses, the mass cancels from both conservation laws, leaving two clean vector statements: the *sum* of the velocities is unchanged, and the *sum of the squared speeds* is unchanged. Comparing $|\mathbf{a}+\mathbf{b}|^2$ with $|\mathbf{a}|^2+|\mathbf{b}|^2$ isolates the scalar product $\mathbf{a}\cdot\mathbf{b}$, and that single quantity drives every part of the question. In the last part, momentum fixes the first ball's velocity once we know the second's, so the energy equation becomes a quadratic in one unknown speed, and a quadratic with a real root has a non-negative discriminant.

Solution

Model the balls as smooth spheres of equal mass m moving on the smooth table, so that during the collision the only horizontal forces are the ones the balls exert on each other.

Conservation laws and relative speed

Momentum and kinetic energy give

$$m\mathbf{v}_1 + m\mathbf{v}_2 = m\mathbf{v}_1' + m\mathbf{v}_2',$$
$$\tfrac{1}{2}m|\mathbf{v}_1|^2 + \tfrac{1}{2}m|\mathbf{v}_2|^2 = \tfrac{1}{2}m|\mathbf{v}_1'|^2 + \tfrac{1}{2}m|\mathbf{v}_2'|^2,$$

that is,

$$\mathbf{v}_1 + \mathbf{v}_2 = \mathbf{v}_1' + \mathbf{v}_2' \quad (1), \qquad |\mathbf{v}_1|^2 + |\mathbf{v}_2|^2 = |\mathbf{v}_1'|^2 + |\mathbf{v}_2'|^2 \quad (2).$$

Take the scalar product of each side of (1) with itself:

$$|\mathbf{v}_1|^2 + |\mathbf{v}_2|^2 + 2\,\mathbf{v}_1\cdot\mathbf{v}_2 = |\mathbf{v}_1'|^2 + |\mathbf{v}_2'|^2 + 2\,\mathbf{v}_1'\cdot\mathbf{v}_2'.$$

Subtracting (2) leaves

$$\mathbf{v}_1\cdot\mathbf{v}_2 = \mathbf{v}_1'\cdot\mathbf{v}_2'. \qquad (3)$$

Now the squared relative speed is

$$|\mathbf{v}_1 - \mathbf{v}_2|^2 = |\mathbf{v}_1|^2 + |\mathbf{v}_2|^2 - 2\,\mathbf{v}_1\cdot\mathbf{v}_2,$$

and by (2) and (3) each term on the right is the same after the collision as before. So $|\mathbf{v}_1 - \mathbf{v}_2|^2 = |\mathbf{v}_1' - \mathbf{v}_2'|^2$, and since speeds are non-negative, $|\mathbf{v}_1 - \mathbf{v}_2| = |\mathbf{v}_1' - \mathbf{v}_2'|$: the relative speed is conserved.

One ball initially at rest

Say $\mathbf{v}_2 = \mathbf{0}$. Then (3) gives

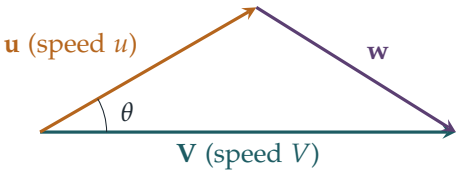
$$\mathbf{v}_1' \cdot \mathbf{v}_2' = \mathbf{v}_1 \cdot \mathbf{0} = 0.$$

Since both balls are moving afterwards, $\mathbf{v}_1' \neq \mathbf{0}$ and $\mathbf{v}_2' \neq \mathbf{0}$, so a zero scalar product means that the two final velocities are perpendicular. (Without that condition the scalar product could vanish only because one velocity is zero, as in a head-on collision, where the balls simply swap velocities.)

The collision that loses energy

Now (1) still holds but (2) doesn't. Call the ball at rest A and the moving ball B. Let B start with velocity $\mathbf{V}$, where $|\mathbf{V}| = V$, and finish with velocity $\mathbf{u}$ of speed u, at angle θ to $\mathbf{V}$. Let A finish with velocity $\mathbf{w}$.

Momentum gives $\mathbf{V} = \mathbf{u} + \mathbf{w}$, so the three velocities form a triangle:



Hence $\mathbf{w} = \mathbf{V} - \mathbf{u}$ and, since $\mathbf{V} \cdot \mathbf{u} = Vu\cos\theta$,

$$|\mathbf{w}|^2 = V^2 - 2Vu\cos\theta + u^2$$

(the cosine rule in the triangle). The final kinetic energy is a fraction $1 - k$ of the initial kinetic energy. Cancelling $\frac{1}{2}m$,

$$u^2 + |\mathbf{w}|^2 = (1-k)V^2 \quad \Longrightarrow \quad 2u^2 - 2Vu\cos\theta + V^2 = (1-k)V^2,$$

so the final speed u of the second ball satisfies

$$\boxed{2u^2 - 2Vu\cos\theta + kV^2 = 0.}$$

Ask yourself The question only asks us to show $k \leq \frac{1}{2}$. Where does an inequality come from, when all we have is an equation?

The speed u is a real number that actually occurs, so this quadratic in u *has* a real root. Its discriminant must therefore be non-negative:

$$4V^2\cos^2\theta - 8kV^2 \geq 0 \quad \Longrightarrow \quad k \leq \tfrac{1}{2}\cos^2\theta \leq \tfrac{1}{2},$$

where we divided by $8V^2 > 0$ (the second ball is moving) and used $\cos^2\theta \leq 1$. So $k \leq \frac{1}{2}$, as required.

Equality needs $\cos^2\theta = 1$ and a double root $u = \frac{1}{2}V\cos\theta$. As $u \geq 0$, this means $\theta = 0$ and $u = \frac{1}{2}V$, and then $\mathbf{w} = \frac{1}{2}\mathbf{V}$ too: the balls move off together at half speed, which is the perfectly inelastic head-on collision. At most half the kinetic energy can be lost.

Checks

- $k = 0$ gives $2u^2 = 2Vu\cos\theta$, so $u = V\cos\theta$ (or $u = 0$). Then $\mathbf{w} = \mathbf{V} - \mathbf{u}$ is perpendicular to $\mathbf{u}$, since $\mathbf{u}\cdot(\mathbf{V} - \mathbf{u}) = Vu\cos\theta - u^2 = 0$. This agrees with the second part.
- A head-on collision with coefficient of restitution e gives $k = \frac{1}{2}(1 - e^2)$, which lies between 0 and $\frac{1}{2}$. With $e = 0.3$ this is $k = 0.455$.
- Numerically, we simulated 10^5 random elastic collisions of equal smooth spheres. Momentum, kinetic energy, $\mathbf{v}_1\cdot\mathbf{v}_2$ and the relative speed all agreed to within 10^{-13}. In 2×10^5 random energy-losing outcomes the quadratic held to 10^{-14}, and the largest k found was 0.49999.

Common slips

- Treating "relative speed" as $\big||\mathbf{v}_1| - |\mathbf{v}_2|\big|$. It is the magnitude of the relative *velocity*, $|\mathbf{v}_1 - \mathbf{v}_2|$.
- In the second part, forgetting to use the fact that both balls are moving. A zero scalar product means perpendicular only if neither vector is zero.
- In the last part, using (2) or (3) again. They came from conservation of kinetic energy, which no longer holds. Only momentum survives.

Beyond STEP

The scalar-product argument is a frame-free version of a standard trick: in the zero-momentum frame, an elastic collision simply rotates the two velocities, keeping their sizes. The maximum loss of $\frac{1}{2}$ is the general fact that only the kinetic energy relative to the centre of mass can be lost, here $\frac{1}{2}\cdot\frac{1}{2}mV^2$. These ideas lead on to scattering theory and to the collisions of special relativity. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers collisions in the centre-of-mass frame carefully.

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